Abstract

Dynamic hedging of options under stochastic volatility remains a fundamentally challenging problem: market incompleteness means volatility risk cannot be perfectly replicated using the underlying asset alone, and classical approaches such as Black–Scholes delta hedging — which assume constant volatility — break down as markets transition between calm and turbulent regimes. This thesis addresses that challenge by developing a regime-adaptive deep hedging framework that combines specialised, learning-based hedging policies with a meta-classification model capable of inferring the prevailing market regime and selecting the appropriate policy in real time.

The framework specialises three neural-network hedgers — trained under low-, medium-, and high-volatility environments simulated via the Heston stochastic volatility model — each learning to minimise replication error through direct interaction with regime-specific market dynamics. A gating meta-model, trained on features derived from realised volatility and historical returns, then learns to identify the underlying regime from observable data alone and route each hedging episode to its corresponding specialist, functioning as a mixture-of-experts system.

Evaluated across six Heston parameter configurations spanning the three regimes, each specialised hedger consistently achieves the lowest replication error within its own training regime, confirming that the models successfully internalise regime-specific hedging behaviour. When episodes from all regimes are pooled to simulate realistic, regime-uncertain deployment, the meta-model achieves the strongest aggregate performance (RMSE 0.01046), outperforming both the best individual specialist (0.01122) and the classical Black–Scholes benchmark (0.01430). Under realistic transaction costs, this advantage narrows — a single robust specialist occasionally matches or slightly exceeds the meta-model — revealing a trade-off between adaptability and stability once trading frictions penalise frequent regime switching.

To test generalisation beyond simulation, the trained models were evaluated out-of-distribution on historical price data for ten large-cap U.S. equities and ETFs (2019–2021), a period spanning the COVID-19 volatility shock. Despite being trained entirely on synthetic Heston paths, the learning-based hedgers transferred effectively to real markets, outperforming a Black–Scholes EWMA benchmark in both frictionless and cost-aware settings, with the meta-model maintaining competitive, stable performance throughout the volatility spike of early 2020.

These findings suggest that decomposing the hedging problem into regime identification and specialised policy learning offers a practical, generalisable path toward more robust derivative hedging under model uncertainty — bridging classical quantitative finance with modern machine learning to address a problem where volatility, by its nature, refuses to sit still.